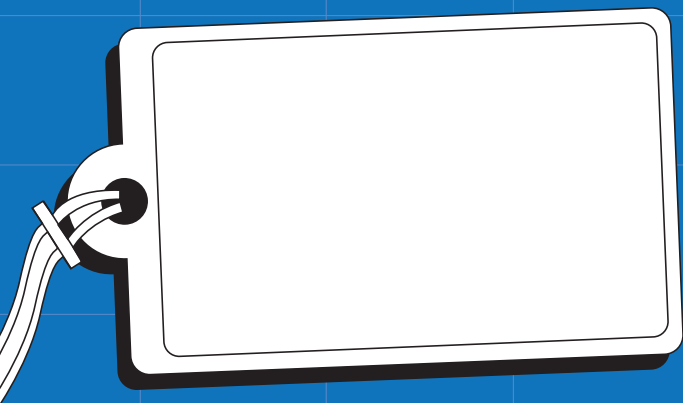




straight
lines

STRAIGHT LINES

Series **K**



MATHLETICS

www.mathletics.com.au

STRAIGHT LINES

Every straight line is related to a linear equation $y = mx + c$ and vice versa. This relationship allows us to find the equation from a line, or a line from an equation. There are also interesting properties of parallel or perpendicular lines.

Try to answer these questions *before* working through this unit.

I used to think:

What do the values of m and c in the equation $y = mx + c$ represent?

Why can the gradient of a straight line be found from $m = \frac{\text{y-intercept}}{\text{x-intercept}}$?

How are the gradients of parallel and perpendicular lines related?

Answer these questions, *after* working through the chapter.

But now I think:

What do the values of m and c in the equation $y = mx + c$ represent?

Why can the gradient of a straight line be found from $m = \frac{\text{y-intercept}}{\text{x-intercept}}$?

How are the gradients of parallel and perpendicular lines related?

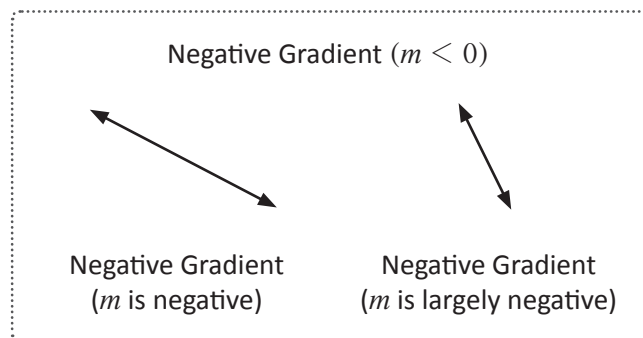
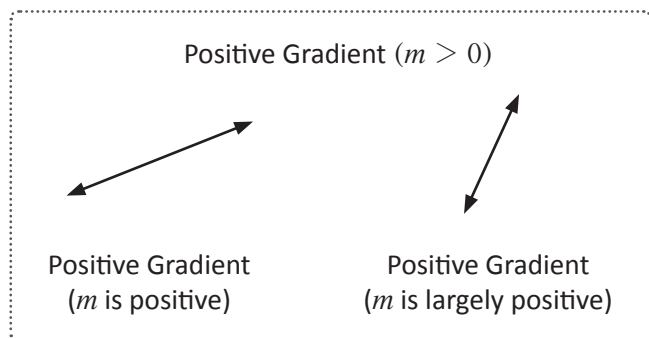


What do I know now that I didn't know before?



Remembering Gradient

Every straight line has a **gradient**, which measures how steep the line is. The greater the gradient, the steeper the line. The symbol of gradient is m . The line leans different ways depending on whether the gradient (or 'slope') is positive or negative.



Gradient is defined as

$$m = \frac{\text{vertical rise}}{\text{horizontal run}}$$

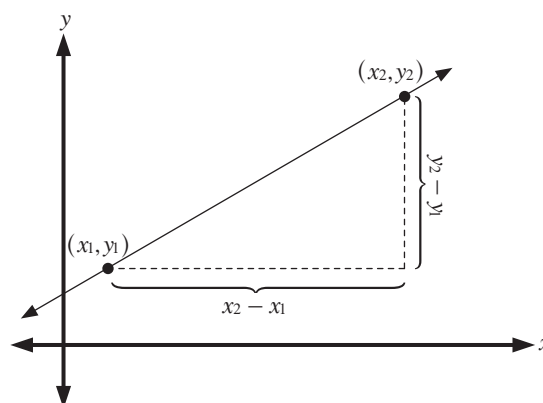
The gradient of any line can be found in two ways:

Method 1

Pick any two points (x_1, y_1) and (x_2, y_2)

$$m = \frac{\text{vertical rise}}{\text{horizontal run}}$$

$$\therefore m = \frac{y_2 - y_1}{x_2 - x_1}$$



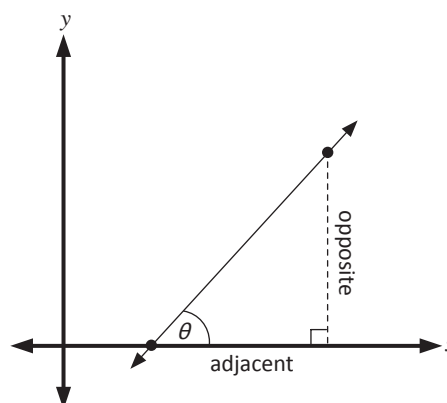
Method 2

Find the gradient from the angle made by the line

$$m = \frac{\text{vertical rise}}{\text{horizontal run}}$$

$$\therefore m = \frac{\text{opposite}}{\text{adjacent}}$$

$$\therefore m = \tan \theta$$



θ is called the **angle of inclination**. It is the angle between the line and positive x -axis.

Gradient-Intercept Form

Every straight line corresponds to an algebraic equation. This can be written like this:

$$\begin{array}{c} \boxed{\text{Gradient}} \quad \downarrow \\ y = mx + c \quad \leftarrow \boxed{\text{y-intercept}} \end{array}$$

"Gradient-Intercept" form depends on the gradient and the y-intercept. y is always the subject of the equation.

Answer these questions involving Gradient-Intercept form

- a** What is the gradient and y-intercept of the straight line corresponding to $y = 2x + 3$

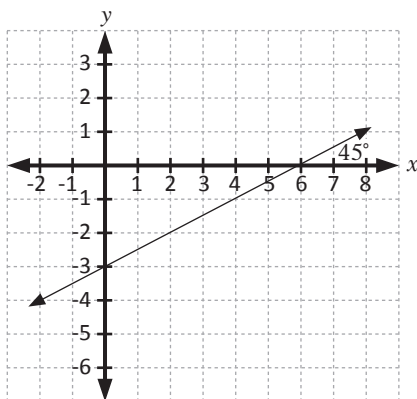
The equation $y = -2x + 3$ is in Gradient-Intercept form.

$$\begin{array}{c} \uparrow \quad \uparrow \\ \boxed{\text{gradient}} \quad \boxed{\text{y-intercept}} \end{array}$$

The gradient is $m = -2$

The y-intercept is $c = 3$

- b** Write the equation of this line in Gradient-Intercept form:



Since $\theta = 45^\circ$ the gradient is

$$\begin{aligned} m &= \tan \theta \\ &= \tan 45^\circ \\ &= \frac{1}{2} \end{aligned}$$

The y-intercept is $c = -3$

The equation of the line is $y = mx + c$

$$y = \frac{1}{2}x - 3$$

General Form of a Line

If all the terms are moved to the left side of the equation and the coefficient of x is NOT negative, then the equation is in General Form. Also, all the coefficients must be integers.

$$\begin{array}{c} \boxed{\text{Not negative}} \quad \rightarrow \quad \begin{array}{c} \downarrow \quad \downarrow \quad \downarrow \\ ax + by + c = 0 \end{array} \quad \boxed{\text{Must be integers}} \end{array}$$

Write $4y = -7x + 8$ in **a** Gradient-Intercept Form and **b** General Form.

- a** Standard form: Make y the subject:

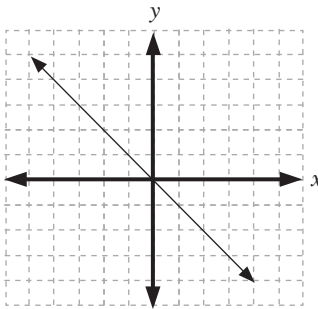
$$y = -\frac{7}{4}x + 2$$

- b** General Form: Move all the terms to one side

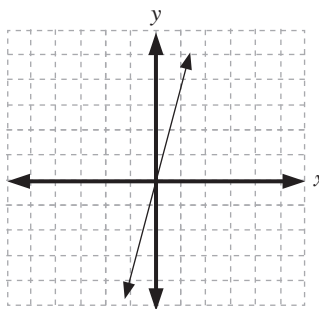
$$7x + 4y - 8 = 0$$

Match the gradients to the graphs below them:

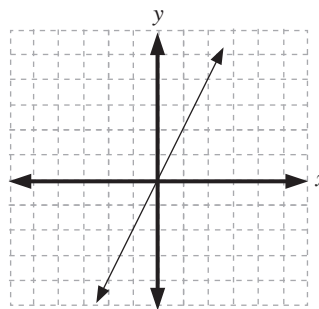
a $m = 2$



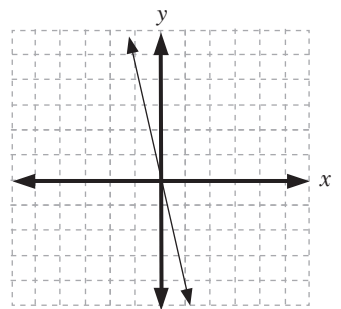
b $m = -1$



c $m = -5$

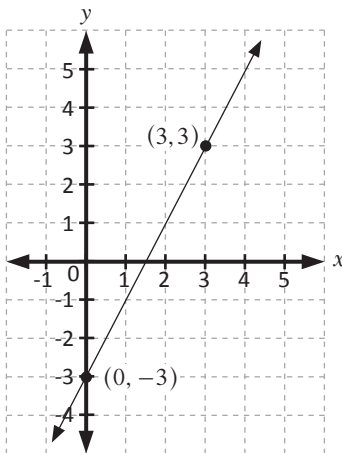


d $m = 4$

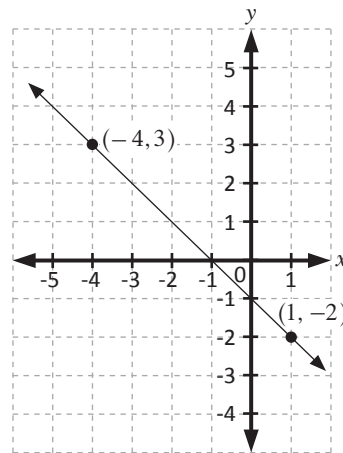


Find the gradient of these lines:

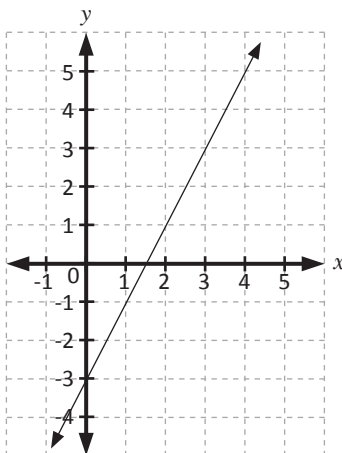
a



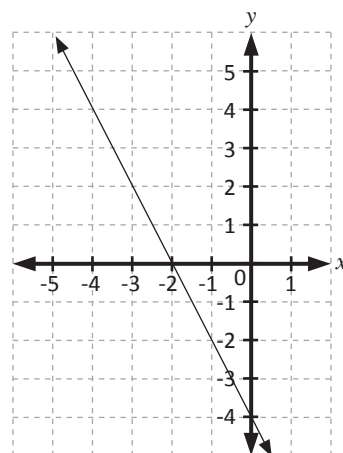
b



c



d



Find the gradient of a line which:

- a Goes through the points $(1, 3)$ and $(-1, -1)$
- b Has angle of inclination $\theta = 0^\circ$
- c Goes through the points $(-2, 8)$ and $(3, -7)$
- d An angle of inclination $\theta = 63.434^\circ$ (nearest unit)

What angle does the line make with the positive x -axis if it has a gradient of 3. (Answer to nearest degree)

If a line has a gradient $m = 3$ and a y -intercept of $c = -1$ then write the corresponding equation in:

- a Gradient-Intercept form
- b General form

If a line has a y -intercept of $c = 5$ and gradient $m = -6$ then write the corresponding equation in:

- a Gradient-Intercept form
- b General form

Points A , B and C are colinear (on the same line) if the gradient of AB is equal to the gradient of BC .
Are the following points colinear?

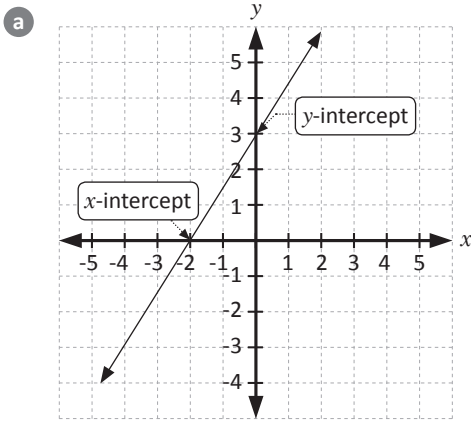
a $A(-1, -5)$, $B(2, 7)$ and $C(-3, -13)$

b $A(-2, 4)$, $B(0, 3)$ and $C(3, -1)$

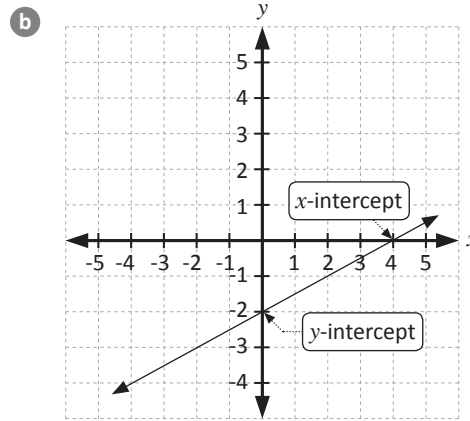
Intercepts

Intercepts are the points where a line cuts the axes. Here are some examples:

Find the intercepts of these lines:



The x -intercept is -2
The y -intercept is 3



The x -intercept is 4
The y -intercept is -2

The intercepts of a line are used to graph a line from an equation. The intercepts can be found from the equation like this:

Find the intercepts of a straight line with these equations:

a $y = 2x - 8$

$$\begin{aligned} 0 &= 2x - 8 \\ 2x &= 8 \\ \therefore x &= 4 \end{aligned}$$

The x -intercept is 4

$$\begin{aligned} y &= 2(0) - 8 \\ &= -8 \end{aligned}$$

The y -intercept is -8

b $3x - y + 9 = 0$

$$y = 3x + 9$$

Rewrite in Gradient-Intercept form

$$\begin{aligned} 0 &= 3x + 9 \\ 3x &= -9 \\ \therefore x &= -3 \end{aligned}$$

The x -intercept is -3

$$\begin{aligned} y &= 3(0) + 9 \\ &= 9 \end{aligned}$$

The y -intercept is 9

As a shortcut, the y -intercept of a line in Gradient-Intercept form, $y = mx + c$, is the value of c .

This page shows how to find the intercepts from an equation. The next page shows how to find the equation from intercepts.

Finding an equation from Intercepts

The equation of a line $y = mx + c$ simply required values for m and c . These values can be found from the intercepts.

$$m = -\frac{y\text{-intercept}}{x\text{-intercept}}$$

$$c = y\text{-intercept}$$

Find the equation of a line with these intercepts

- a** x -intercept -3 , y -intercept 12

$$m = -\frac{y\text{-intercept}}{x\text{-intercept}} = -\left(\frac{12}{-3}\right) = 4$$

$$c = y\text{-intercept}$$

$$= 12$$

$$y = 4x + 12$$

- b** x -intercept 2 , y -intercept 5

$$m = -\frac{y\text{-intercept}}{x\text{-intercept}} = -\left(\frac{5}{2}\right) = -\frac{5}{2}$$

$$c = y\text{-intercept}$$

$$= 5$$

$$y = -\frac{5}{2}x + 5$$

This can be converted to General form if necessary:

a $4x - y + 12 = 0$

b $-\frac{5}{2}x - y + 5 = 0$

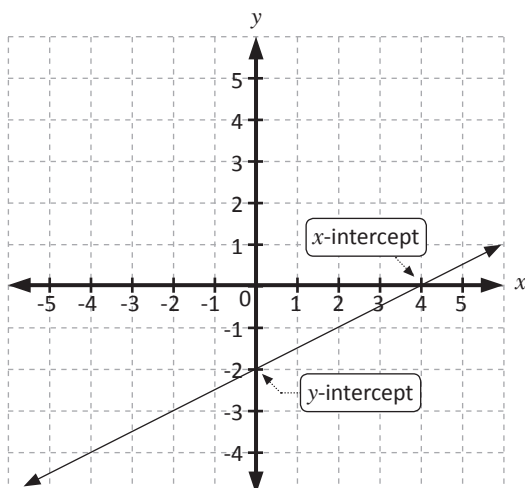
$$\therefore \frac{5}{2}x + y - 5 = 0$$

$$\therefore 5x + 2y - 10 = 0$$

Remember, in General form the coefficient of x must be positive and all the coefficients must be integers

The intercepts can be used to find the equation of a line. Here is an example:

Find the equations of the following line in Gradient-Intercept form:



- Find the intercepts from the graph
The x -intercept is 4 . The y -intercept is -2

- Use the intercepts to find the gradient

$$m = -\frac{y\text{-intercept}}{x\text{-intercept}} = -\left(\frac{-2}{4}\right) = \frac{1}{2}$$

- Use the intercepts to find the value for c
 $c = y\text{-intercept} = -2$

$$\therefore \text{the equation of the line is } y = \frac{1}{2}x - 2$$

Using an Equation to Draw a Line

The intercepts can also be used to draw the line from an equation. This is done in two steps.

Draw the straight line corresponding to the equation $y = -3x - 6$

Step 1: Find the intercepts

$$0 = -3x - 6$$

$$3x = -6$$

$$\therefore x = -2$$

The x -intercept is -2

To find the x -intercept,
make $y=0$ and solve for x

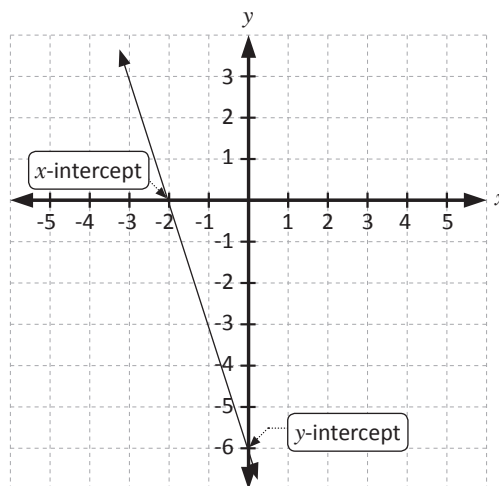
$$y = -3(0) - 6$$

$$= -6$$

The y -intercept is -6

To find the y -intercept,
make $x=0$ and solve for y

Step 2: Draw a line through the intercepts



If the line is in General form, then convert it to Gradient-Intercept form first.

Draw the straight line corresponding to the equation $4x + 2y - 8 = 0$

Convert the equation to Gradient intercept form:

$$4x + 2y - 8 = 0$$

$$\therefore y = -2x + 4$$

Step 1: Find the intercepts

$$0 = -2x + 4$$

$$2x = 4$$

$$\therefore x = 2$$

The x -intercept is 2

To find the x -intercept,
make $y=0$ and solve for x

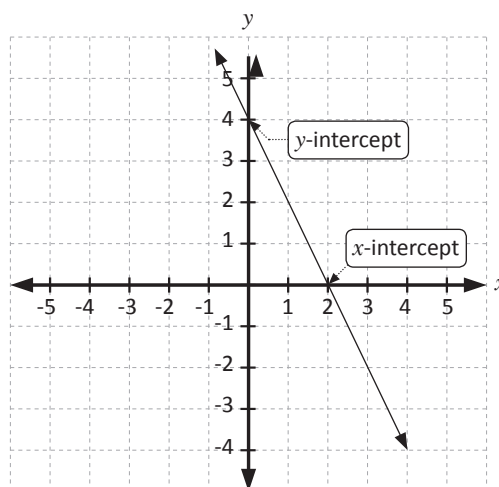
$$y = -2(0) + 4$$

$$= 4$$

The y -intercept is 4

To find the y -intercept,
make $x=0$ and solve for y

Step 2: Draw a line through the intercepts



Find the intercepts of these lines:

a $y = -3x + 6$

b $y = -4x - 8$

c $5x - 2y + 10 = 0$

d $3x + 7y + 21 = 0$

e $y = -\frac{1}{4}x + 1$

f $3x - 5y - 15 = 0$

g $y = -\frac{3}{2}x - 3$

h $5x + 6y - 30 = 0$

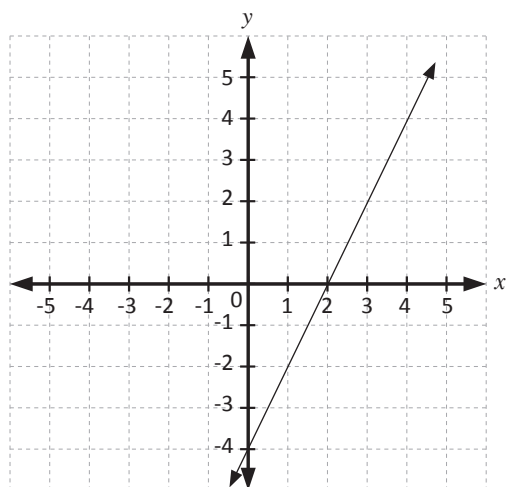
Find the equation of a line which has intercepts, in Gradient-Intercept Form:

- a x -intercept is 1, y -intercept is 3
- b x -intercept is 7, y -intercept is 14
- c x -intercept is -4 , y -intercept is 2
- d x -intercept is -3 , y -intercept is -15

Find the equation of a line which has intercepts, in General Form:

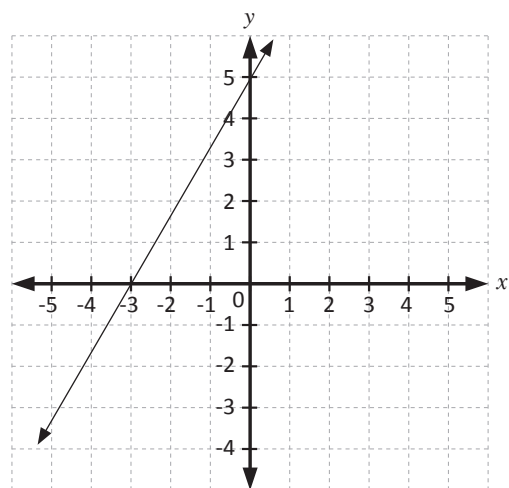
- a x -intercept is 4, y -intercept is 3
- b x -intercept is -2 , y -intercept is 7
- c x -intercept is -6 , y -intercept is 4
- d x -intercept is -4 , y -intercept is -5

Answer the following questions about this straight line:



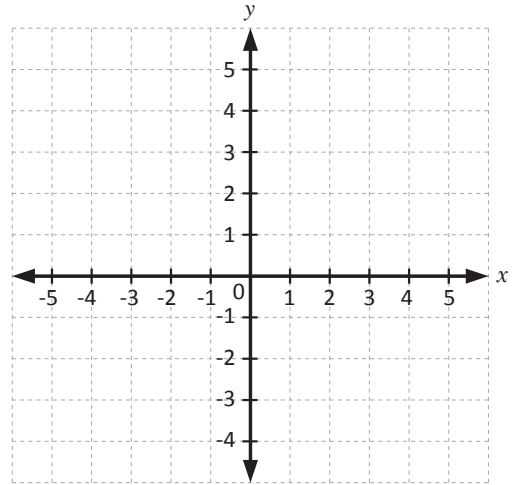
- What are the intercepts of this line?
- Will the gradient be positive or negative?
- Find the gradient of the line
- Write the equation of this line in Gradient-Intercept form
- Write the equation of this line in General form

Write the equation of the following line in General form

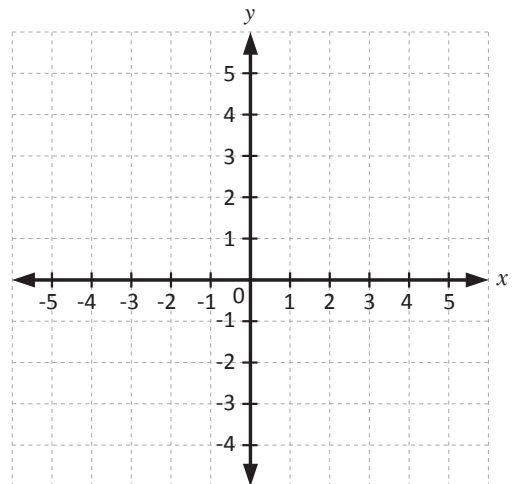


Use the axes below to draw straight lines from the following equations:

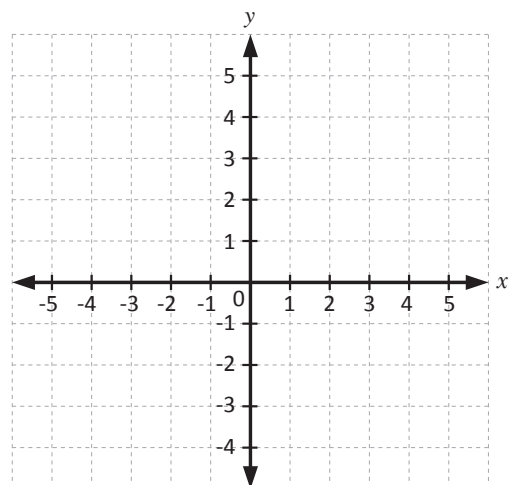
a $y = x - 2$



b $y - 4x = 4$



c $7x - 4y - 28 = 0$



Finding an Equation from any Two Points

In the equation $y = mx + c$, only values for m and c need to be found for the equation of the line.

If a line passes through the two points (x_1, y_1) and (x_2, y_2) , then:

- The value for m (the gradient) can be found using the formula

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{or} \quad m = \frac{y_1 - y_2}{x_1 - x_2}$$

- The value for c is found by substituting either of the points into the equation of the straight line.

Find the equation of the line which passes through $A(1, -1)$ and $B(2, 3)$

x_1 y_1 x_2 y_2

Use the formula for gradient to find m

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{3 - (-1)}{2 - 1}$$

$$\therefore m = 4$$

$$\therefore y = 4x + c$$

If B was chosen as (x_1, y_1) and A was chosen as (x_2, y_2) , then m would still be the same.

Substitute a point into the above equation and solve for c . Substituting $A(1, -1)$ point gives:

$$-1 = 4(1) + c$$

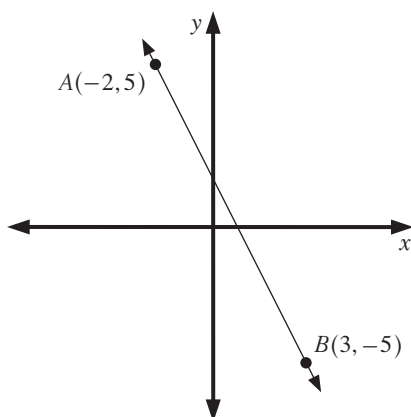
$$\therefore c = -5$$

Either A or B can be substituted here. The answer will always work out.

The equation of the line is $y = 4x - 5$

Sometimes the graph needs to be used to find the points.

Find the equation of the following line:



- The line goes through points $A(-2, 5)$ and $B(3, -5)$
- Use the formula for gradient to find m

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-5 - 5}{3 - (-2)}$$

$$\therefore m = -2$$

- Substitute point A (or B) into the equation $y = -2x + c$

$$5 = -2(-2) + c$$

$$\therefore c = 1$$

Finding an Equation from a Point and Gradient

In the equation $y = mx + c$, only values for m and c need to be found for the equation of the line. If the gradient of the line is given, then there is no need to find m . All that is required is to find c .

To find c , substitute the point into the equation.

Find the equation of a line which goes through the point $A(3, -4)$ and has a gradient of -3

From the question $m = -3$

$$\therefore y = -3x + c$$

Substitute point $A(3, -4)$ into this equation

$$-4 = -3(3) + c$$

$$\therefore c = 5$$

$\therefore$ The equation of the line is $y = -3x + 5$

Here is an example using both methods

Find the equations of the lines passing through AB and CD below, if CD has a gradient of -2

For the line passing through AB

- The line goes through points $A(-4, -22)$ and $B(3, 6)$
- Use the formula for gradient to find m

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-6(-22)}{3 - (-4)}$$

$$\therefore m = 4$$

- Substitute point B (or A) into the equation $y = 4x + c$

$$6 = 4(3) + c$$

$$\therefore c = -6$$

$\therefore$ The line passing through AB had equation $y = 4x - 6$

For the line passing through CD

From the question $m = -2$.

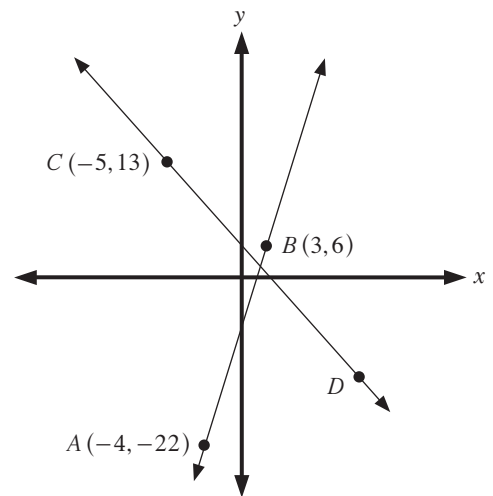
$$\therefore y = -2x + c$$

Substitute point $C(-5, 13)$ into this equation

$$13 = -2(-5) + c$$

$$\therefore c = 3$$

$\therefore$ The equation of the line is $y = -2x + 3$



Find the equations of lines which pass through these points:

a $A(2, 1)$ and $B(3, 2)$

b $C(2, -1)$ and $D(4, -5)$

c $P(4, 11)$ and $Q(-2, -13)$

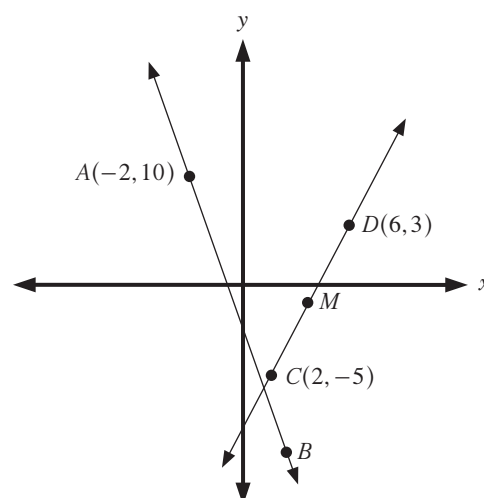
d $A(-3, 16)$ and $B(-1, 10)$

Find the equations of these lines. Write the equation in Gradient-Intercept form and General form

- a A line passing through $A(-1, -6)$ with $m = 4$
- b A line with a gradient of -1 passing through $B(-3, 6)$
- c A line passing through $C(6, 1)$ with $m = \frac{2}{3}$
- d A line with a gradient of $-\frac{2}{5}$ and passing through $D(15, -9)$

Use the graph on the right to answer these questions below

- a What is the equation of line AB if it has gradient $m = -4$



- b What is the equation of the line passing through points C and D ?

- c What are the coordinates of M , the midpoint of C and D ?

- d What would the equation of a line passing through AM be?

If the intercepts of a straight line are known, then we know it passes through the points:

$(x\text{-intercept}, 0)$ and $(0, y\text{-intercept})$

- a Find the gradient of the line passing through these two points.

- b Where have you used this before?

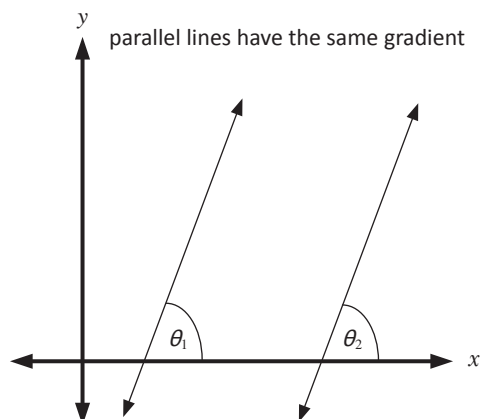
Parallel Lines

Parallel lines make the same angle with the x -axis.

In the diagram, the lines are parallel. This means $\theta_1 = \theta_2$.

$$\therefore \tan \theta_1 = \tan \theta_2$$

Since the gradient of a line is $m = \tan \theta$, this means that parallel lines have the same gradient.



Which of the following lines are parallel?

- Line 1: $y - 3x = 1$
- Line 2: $2y + 2 = 6x$
- Line 3: $y - 6x + 4 = 0$

Rewrite each line in gradient-intercept form:

- Line 1: $y = 3x + 1$
- Line 2: $y = 3x - 1$
- Line 3: $y = 6x - 4$

Line 1 and Line 2 are parallel since they have the same gradient $m = 3$.

Sometimes the fact that lines are parallel can be used to find points on the line.

Four points have the coordinates $A(-2, -11)$, $B(3, 9)$, $C(-2, -4)$ and $D(x, -12)$.

Solve for x so that AB is parallel to CD .

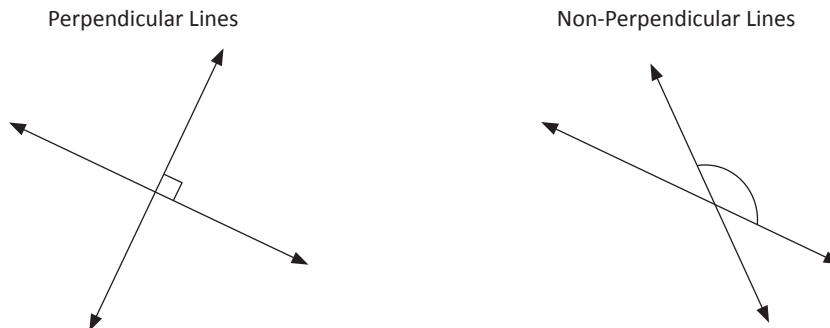
Let m_1 be the gradient of AB and let m_2 be the gradient of CD . Parallel lines have equal gradients so $m_1 = m_2$.

$$\begin{aligned}
 m_1 &\longrightarrow \frac{9 - (-11)}{3 - (-2)} = \frac{-12 - (-4)}{x - (-2)} \longleftarrow m_2 \\
 &4 = \frac{-8}{x + 2} \\
 \therefore x + 2 &= -2 \\
 \therefore x &= -4
 \end{aligned}$$

So D has coordinates $D(-4, -12)$

Perpendicular Lines

Lines are perpendicular if they cut each other at 90° :



Straight lines will be perpendicular if the product of their gradients is -1 . For example:

- A line with a gradient 5 will be perpendicular to a line with a gradient of $-\frac{1}{5}$, since $5 \times -\frac{1}{5} = -1$.
- A line with a gradient of $\frac{1}{2}$ will be perpendicular to a line with a gradient of -2 , since $\frac{1}{2} \times -2 = -1$.
- A line with a gradient of -1 will be perpendicular to a line with a gradient of 1 , since $-1 \times 1 = -1$.

Which of the lines below are perpendicular to each other?

- Line 1: $y = 2x - 3$
- Line 2: $y = \frac{1}{2}x + 4$
- Line 3: $y = -\frac{1}{2}x - 5$

Line 1

= Gradient $\times$ Line 2 Gradient

$$= 2 \times \frac{1}{2}$$

$$= 1$$

$\therefore$ Not Perpendicular

Line 2

= Gradient $\times$ Line 3 Gradient

$$= 2 \times -\frac{1}{2}$$

$$= -1$$

$\therefore$ Perpendicular

Line 3

= Gradient $\times$ Line 3 Gradient

$$= \frac{1}{2} \times -\frac{1}{2}$$

$$= -\frac{1}{4}$$

$\therefore$ Not Perpendicular

Let's say Line 1 has a gradient of m_1 and Line 2 has a gradient of m_2 . If they are perpendicular then

$$m_1 \times m_2 = -1$$

or

$$m_2 = -\left(\frac{1}{m_1}\right)$$

The perpendicular gradient is the negative reciprocal

Remember to make sure that the equations are in **Gradient-Intercept** form before checking for gradients.

Finding Parallel Lines

The gradient can be used to find parallel lines. Remember, only values for m and c are needed for the equation of a line.

Find the equation of a line parallel to $y = 2x + 3$ that goes through the point $(4, 2)$.

- Since the line is parallel to $y = 2x + 3$, it must have the same gradient.
- The gradient of the line is $m = 2$. $\therefore$ the equation of the line is $y = 2x + c$.
- To find c , substitute the given point into the equation:

Substitute $(4, 2)$ into $y = 2x + c$

$$\therefore 2 = 2(4) + c$$

$$\therefore c = -6$$

$\therefore y = 2x - 6$ is the equation of the line parallel to $y = 2x + 3$ that passes through $(4, 2)$

If the equation of the line is not in Gradient-Intercept form, remember to change it first.

Finding Perpendicular Lines

The gradient can be used to find parallel lines too, in the same way.

Find the equation of a line parallel $3x + y + 1 = 0$ to that goes through the point $(-6, 1)$

This line is in General form, so first convert to Gradient-Intercept form: $y = -3x - 1$

- Since the line is perpendicular to $y = -3x - 1$

$$m = -\left(\frac{1}{-3}\right) = \frac{1}{3}$$

- The gradient of the line is $m = \frac{1}{3}$. $\therefore$ the equation of this line is

$$y = \frac{1}{3}x + c$$

- To find c , substitute the given point into the equation:

Substitute $(-6, 1)$ into $y = \frac{1}{3}x + c$

$$\therefore 1 = \frac{1}{3}(-6) + c$$

$$\therefore c = 3$$

$\therefore y = 2x - 6$ is the equation of the line parallel to $y = \frac{1}{3}x + 3$ that passes through $(-6, 1)$

What is the difference between parallel lines and perpendicular lines?

Match the lines on the left to the lines parallel to them on the right

$$y = 2x + 1 \bullet$$

$$\bullet y = 4x + 10$$

$$y = -4x + 4 \bullet$$

$$\bullet 2y + 8x = -6$$

$$3x + y - 2 = 0 \bullet$$

$$\bullet 2x - y - 3 = 0$$

$$4x - y - 1 = 0 \bullet$$

$$\bullet 7 - 2x - y = 0$$

$$y = -2x + 3 \bullet$$

$$\bullet y = -3x$$

Match the lines on the left to the lines perpendicular to them on the right

$$y = x - 7 \bullet$$

$$\bullet y = \frac{1}{2}x$$

$$y = 4x + 3 \bullet$$

$$\bullet y = -\frac{1}{6}x - 4$$

$$y = -2x + 6 \bullet$$

$$\bullet y = -\frac{1}{4}x + 1$$

$$y = 6x + 9 \bullet$$

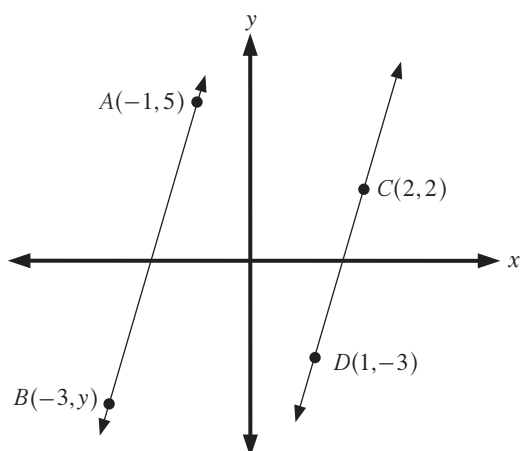
$$\bullet y = -x + 2$$

$$4x + y - 1 = 0 \bullet$$

$$\bullet y = \frac{1}{4}x + 3$$

The two lines below are parallel.

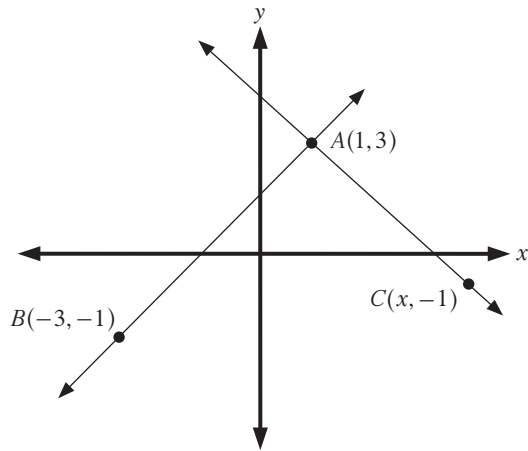
- a Find the value of y in B .



- b Write the equation of each line in Gradient-Intercept and General form.

The two lines below are perpendicular.

- a Find the value of x in C .



- b Write the equation of each line in Gradient-Intercept and General form.

Find the following parallel lines in General form:

- a A line parallel to $y = 3x - 4$ passing through the point $(3, 14)$
- b A line parallel to $4x + y - 7 = 0$ passing through the point $(4, -17)$

Find the following perpendicular lines in General form:

a A line perpendicular to $y = \frac{1}{2}x$ passing through the point $(8, -12)$

b A line parallel to $3x + y - 2 = 0$ passing through the point $(-9, 2)$

Is $y = \frac{1}{4}x - 2$ perpendicular to the line joining $A(-3, 13)$ and $B(2, -7)$?

The line $ax + by + c = 0$ is perpendicular to one of the following lines and parallel to the other:

- $bx - ay = 0$
- $y = -\frac{a}{b}x + 3$

Which line is it perpendicular to, and which line is it parallel to?

